Sang Yoon Byun

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Education

University of California, San Diego

Ph.D. Candidate, Computer Science

Thesis: Computational Framework for End-to-End Variant Analysis of Paralogous Genes
Commitee: Vineet Bafna, Melissa Gymrek, Niema Moshiri, Vikas Bansal
Advisor: Vikas Bansal

M.S., Computer Science

Advisor: Vikas Bansal

Grinnell College

B.A., Computer Science

Advisor: Jerod Weinman
Minor: Policy Studies

Teaching Experience

University of California, San Diego

CSE 29: Systems Programming and Software Tools

Head Teaching Assistant Mar – Jun 2026

· Managed course logistics including grading schedule, exam logistics, and TA/tutor task distribution.
· Lectured one 80-minute class, covering File I/O in C programming, process states, memory hierarchy.
· Led three 1-hour discussion sessions and one 1-hour office hours every week.
· Number of Students: 314 undergraduate students

CSE 599: Teaching Methods in Computer Science

Mentor Teaching Assistant Jan – Mar 2026

· Facilitated and led 50-minute practicum sessions about teaching pedagogy for first-time TAs.
· Number of Students: 5 graduate students

CSE 8A/6R: Introduction to Programming

Teaching Assistant Sep – Dec 2025

· Developed question bank (30+ questions) for midterm and final exam.
· Created additional conceptual and coding problems for students to gain additional practice.
· Led three 1-hour discussion sessions and two 1-hour office hours every week.
· Number of Students: 457 undergraduate students

Grinnell College

FairyBytes: Student-led Programming Club

Undergraduate Teaching Mentor Jan – May 2022

· Provided weekly 1-hour sessions teaching Python programming and data science concepts and skills.
· Led collaborative coding sessions where students built on top of one another to build a snake game.
· Number of Students: 5 undergraduate students

Professional Development

Introduction to College Teaching

UC San Diego Engaged Teaching Hub Sep – Present

· Currently enrolled in the course to further develop teaching pedagogy and strategies.

Workshop on Education for Bioinformatics

International Society for Computational Biology Jul 2026

· Collaborated to define ethical and professional boundaries for AI use in bioinformatics training.
· Developed a shared Golden Prompt repository and a set of best practices and guidelines.

Professional Experience

Nurix Therapeutics, Inc.

Cheminformatics & Machine Learning Intern Jun – Sep 2026

· Developed computational tools to analyze DNA-encoded library (DEL) selection data and identify sequence-driven binding signals.

Allen Institute for Brain Science

Computational Neuroscience Intern Jun – Aug 2022

· Built a Python package that employs dataset alignment methods and semi-supervised clustering techniques to correct non-biologically shifts and reduce noise across large electrophysiology datasets.

Artificial Intelligence Institute of Seoul National University

Computational Immunology Intern Jun – Aug 2021

· Designed a method that applies NLP-based deep embedding techniques to protein sequences and classifies B-cell receptor repertoires of COVID-19 patients and healthy subjects with over 87% accuracy.

Los Alamos National Laboratory

Parallel Computing Research Intern Jun – Aug 2020

· Optimized high-performance computational fluid dynamics code by porting Fortran code to C/C++.

Honors and Awards

Awards

Doctoral Award for Excellence in Teaching

University of California, San Diego La Jolla, CA
Jun 2026

Intelligent Systems for Molecular Biology Conference Fellowship

International Society for Computational Biology La Jolla, CA
May 2026

Publications

Peer-Reviewed Journal Articles

Aggregate variant calling using short reads enables population and disease studies for paralogous genes

Timothy Prodanov, Sang Yoon Byun, and Vikas Bansal Sep 2026

Human Genetics and Genomics Advances
`https://doi.org/10.1016/j.xhgg.2026.100653`

Likelihood-based optimization enables accurate copy number estimation for paralogous genes using exome data

Sang Yoon Byun and Vikas Bansal Jul 2026

Bioinformatics, Oxford University Press
`https://doi.org/10.1093/bioinformatics/btag221`

Preprints

Computational analysis of B cell receptor repertoires in COVID-19 patients using deep embedded representations of protein sequences

Inyoung Kim, Sang Yoon Byun, Sangyeup Kim, Sangyoon Choi, Jinsung Noh, Junho Chung, Byung Gee Kim
`https://www.biorxiv.org/content/10.1101/2021.08.02.454701v3` Nov 2021

Presentations

Oral Presentations

Likelihood-based optimization for copy number estimation of paralogous genes

Intelligent Systems for Molecular Biology Conference Washington, DC
Jul 2026

Accurate estimation of copy number for duplicated genes using whole-exome sequencing

Research in Computational Molecular Biology Conference Seoul, South Korea
Apr 2025

CD3 B-cell receptor repertoires as deep embedded representation of protein sequences

Artificial Intelligence Institute at Seoul National University Seoul, South Korea
Jun 2021

Poster Presentations

Likelihood-based optimization for copy number estimation of paralogous genes

UC San Diego: Jacob School of Engineering's 44th Annual Research Expo La Jolla, CA
Apr 2026

Dataset alignment and integration: Exploring electrophysiology shifts in acute versus culture interneurons and human neocortical layer 1 cells

Allen Institute for Brain Science: Summer Intern Symposium Seattle, WA
Aug 2022

Lagrangian Particle Models in HIGRAD/FIRETEC and FIESTA

Los Alamos National Laboratory: Parallel Computing Summer Research Institute Los Alamos, NM
Aug 2020

Service

Mentoring

Senior TA Panelist

UCSD CSE 599: Teaching Methods in Computer Science Oct 2026

· Led group discussion on teaching strategies, including effective time management as a PhD student and how to best manage use of GenAI by students.

Graduate Student Panelist

UC San Diego Warren College Honors Council Feb 2026

· Invited to share my experiences in graduate school, including advice on graduate admissions, transitioning from undergraduate to graduate study, coursework and research expectations.

International Pre-orientation Program Mentor

Grinnell College Apr – Aug 2019

· Mentored a group of eight international, undergraduate students and actively reached out to individuals who were struggling academically and/or culturally.

· Organized various events for international students throughout the Fall 2019 semester.

Consulting

Venture Fellow

UC San Diego Rady School of Management Apr 2024 – Jun 2025

· Led competitive analysis for a spatial transcriptomics startup, identifying strategic partnership opportunities with pharmaceutical companies and research hospitals.

· Developed pro forma financial projections for a agricultural technology startup, leveraging airborne eDNA metabarcoding technologies to monitor on-farm pest populations.

· Navigated regulatory pathways and interviewed KOLs to refine beachhead market and value proposition for a med-tech startup developing a non-invasive diagnostic for predicting sperm-retrieval success.

Volunteer Full Stack Developer

Develop for Good Apr – Jun 2021

· Worked on brownfield project, establishing check in/out service for *Vergil*, a web app designed to help probationers and parolees navigate their case plan responsibilities.